

# REVENUE DECLINE INVESTIGATION

Why Revenue declined by 43%?

This project investigates the decline across two areas: Product and Geography. A drill-down approach is used to identify the root causes and understand the key contributors to the decline. Additionally, Financial Impact and Recovery Analysis were conducted to assess the overall business performance.

The goal is to identify the contributing factors, understand the financial impact of the decline, and pinpoint areas of recovery, so the business can understand what led to the decline.

## Overview

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- 01 Executive Overview
- 02 Root-cause Analysis
- 03 Financial Impact
- 04 Recovery Analysis
- 05 Recommendation
- 06 Additional Information (ERD | Project Workflow | Analytical Workflow)

## About this Project

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**TOOLS** SQL | Power BI | MS-Excel

**LAST UPDATED** Aug 2026

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# Executive Overview

Introduction

Executive Overview

Root-cause Analysis

Financial Impact

Recovery Analysis

Recommendations

Increase

Decrease

Neutral

Select Category

All

Select Subcategory

All

Select Continent

All

Select Country

All

Clear all slicers

Revenue %YoY

-43.1% ▼

Revenue 2011

\$152.2M

Revenue 2012

\$86.6M

Revenue Change

-\$65.6M

Gross Profit %YoY

-43.7% ▼

Gross Profit 2011

\$89.3M

Gross Profit 2012

\$50.3M

Gross Profit Change

-\$39.0M

Quantity Sold %YoY

-41.9% ▼

Quantity Sold 2011

442,660

Quantity Sold 2012

257,330

Quantity Sold Change

-185,330

Avg. Selling Price %YoY

-2.2% ▼

ASP 2011

\$343.8

ASP 2012

\$336.4

ASP Change

-\$7.4

Revenue Trend Analysis

2011 2012

Revenue

Gross Profit

Quantity Sold

ASP

0.5M

Apr

May

Jun

Jul

Aug

Sep

Oct

0.0M



Apr, May, June, July and October are selected for further investigation based on their contribution to the Revenue and Quantity Sold decline.

Revenue Recovery

Apr

-46.3%

May

-47.6%

Jun

-49.2%

Jul

-43.5%

Aug

5.6%

Sep

3.4%

Oct

-40.2%

-40%

-20%

0%

Monthly Revenue Contribution %YoY

Month

%Revenue

%Gross Profit

%Quantity Sold

%Contribution by Month

Apr

-46.3%

-47.1%

-42.7%

-19.9%

May

-47.6%

-48.1%

-47.0%

-27.6%

Jun

-49.2%

-49.7%

-48.0%

-27.5%

Jul

-43.5%

-44.5%

-42.8%

-4.3%

Aug

5.6%

5.4%

15.8%

0.9%

Sep

3.4%

1.9%

19.2%

1.1%

Oct

-40.2%

-40.4%

-41.9%

-22.8%

Total

-43.1%

-43.7%

-41.9%

-100.0%

Root-Cause Analysis

Quantity Sold Change  
-185K

Bottom 1 Class  
103%  
% of Decline

Bottom 5 Categories  
98.8%  
% of Decline

Bottom 9 Subcategories  
86.7%  
% of Decline

Bottom 4 Manufacturers  
79%  
% of Decline

Product

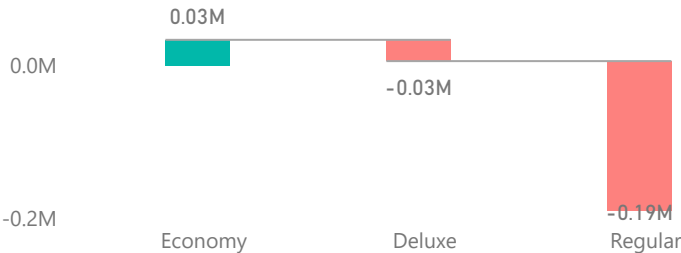
Geography

Each chart below drills into the previous one: Class to Category, Category to Subcategory, and Subcategory to Manufacturer, using the bottom performing segments at each level.

Quantity Change by Class

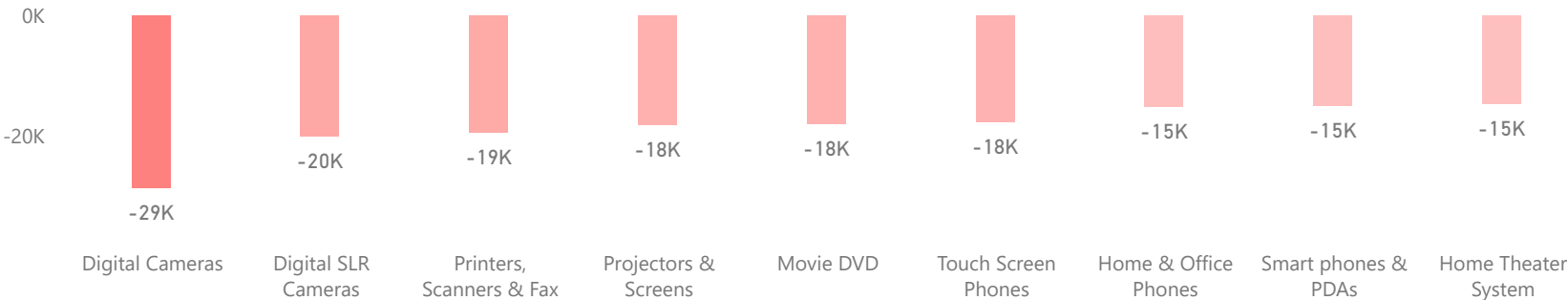
(within Apr, May, Jun, Jul & Oct)

IncreaseDecrease



Quantity Change of Bottom 9 Subcategories

(within previous 5 Categories)



Category Performance by Quantity Change

(within Regular Class)

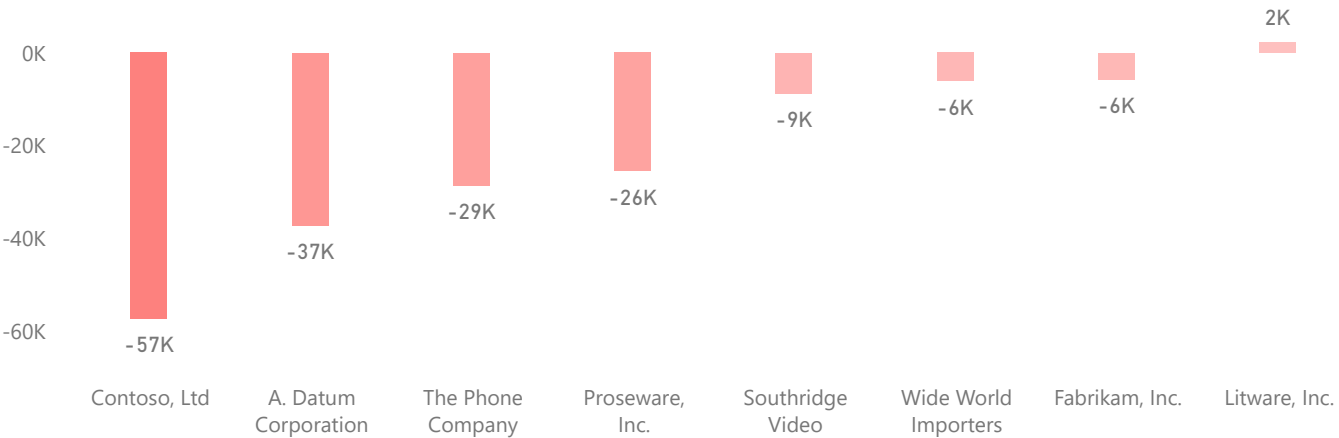
Details

Visual

| Product_Category            | Quantity Change | %YoY   | %Contribution by Category |
|-----------------------------|-----------------|--------|---------------------------|
| Audio                       | -4,340          | -58.6% | -2.3%                     |
| Cameras & Camcorders        | -55,730         | -51.0% | -30.1%                    |
| Cell Phones                 | -51,500         | -51.1% | -27.8%                    |
| Computers                   | -31,120         | -24.9% | -16.8%                    |
| Music, Movies & Audio Books | -23,020         | -52.8% | -12.4%                    |
| TV & Video                  | -19,620         | -34.6% | -10.6%                    |
| Total                       | -185,330        | -41.9% | -100.0%                   |

Quantity Change by Manufacturer

(within previous 9 Subcategories)



Financial Impact

Revenue %YoY-43.1% ▼

|                |          |
|----------------|----------|
| Revenue 2011   | \$152.2M |
| Revenue 2012   | \$86.6M  |
| Revenue Change | -\$65.6M |

Gross Profit %YoY-43.7% ▼

|               |          |
|---------------|----------|
| Profit 2011   | \$152.2M |
| Profit 2012   | \$86.6M  |
| Profit Change | -\$65.6M |

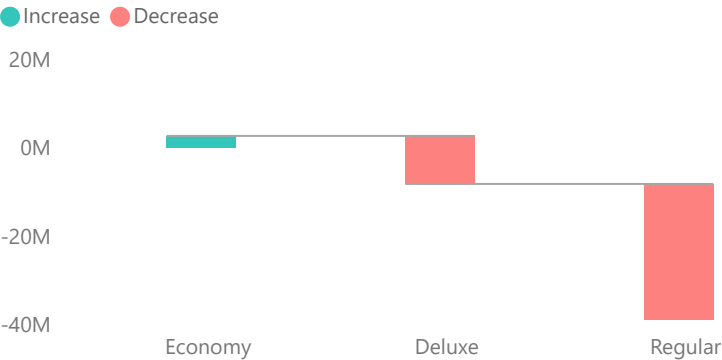
Cost %YoY-42.4% ▼

|             |          |
|-------------|----------|
| Cost 2011   | \$62.9M  |
| Cost 2012   | \$36.3M  |
| Cost Change | -\$26.7M |

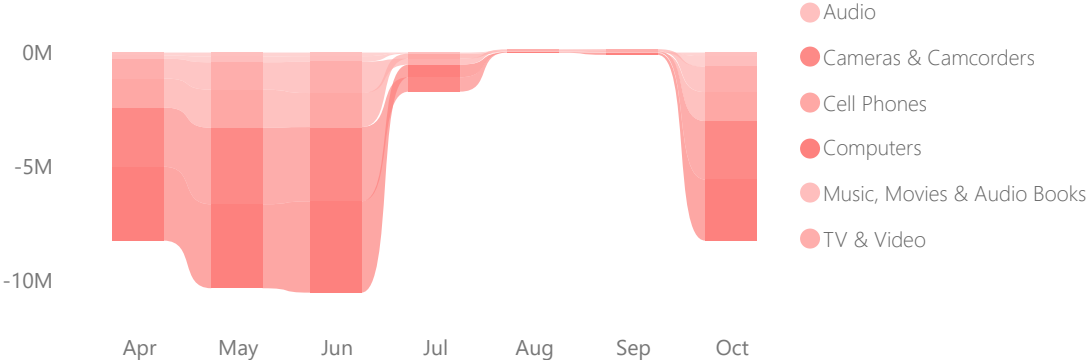
Profit Margin %YoY-42.6% ▼

|                      |         |
|----------------------|---------|
| Profit Margin 2011   | \$2.5M  |
| Profit Margin 2012   | \$1.4M  |
| Profit Margin Change | -\$1.1M |

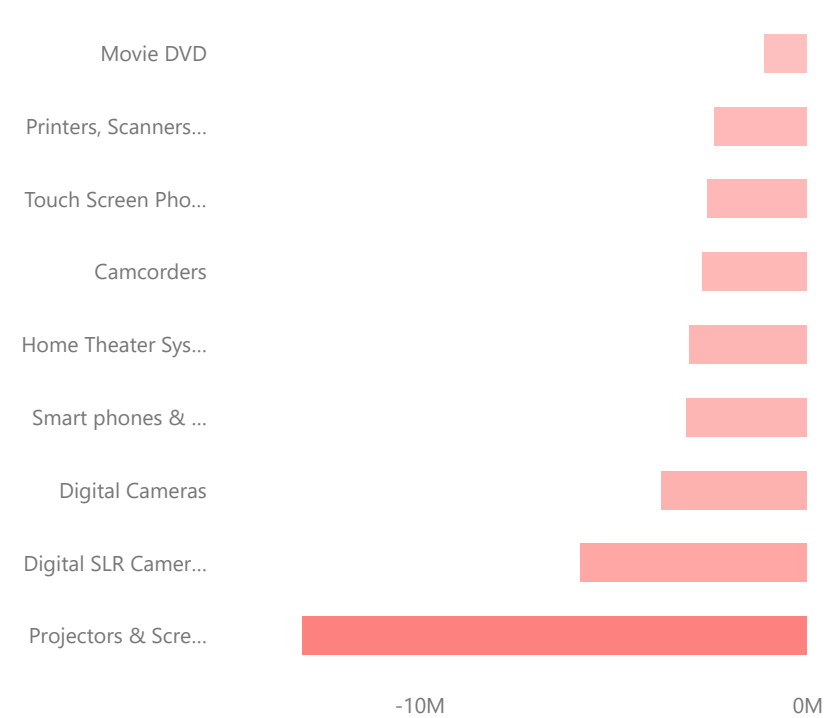
Profit Change by Product Class



Profit Change Trend by Product Category



Profit Change by Bottom 9 Product Subcategories



Profit Change by Country



# Recovery Analysis

Product

Geography & Store



Each chart below drills into the previous one: Class to Category, Category to Subcategory and Subcategory to Manufacturer, using the top performing segment at each level.

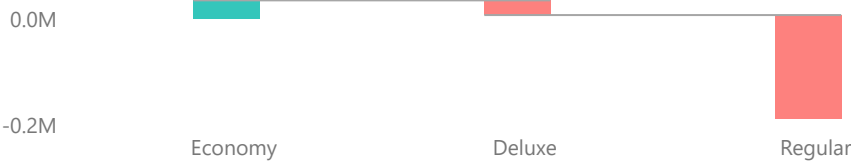
Revenue %YoY -79.9% ▼

|                |          |
|----------------|----------|
| Revenue 2011   | \$148.1M |
| Revenue 2012   | \$82.3M  |
| Revenue Change | -\$65.8M |

Quantity Change by Product Class

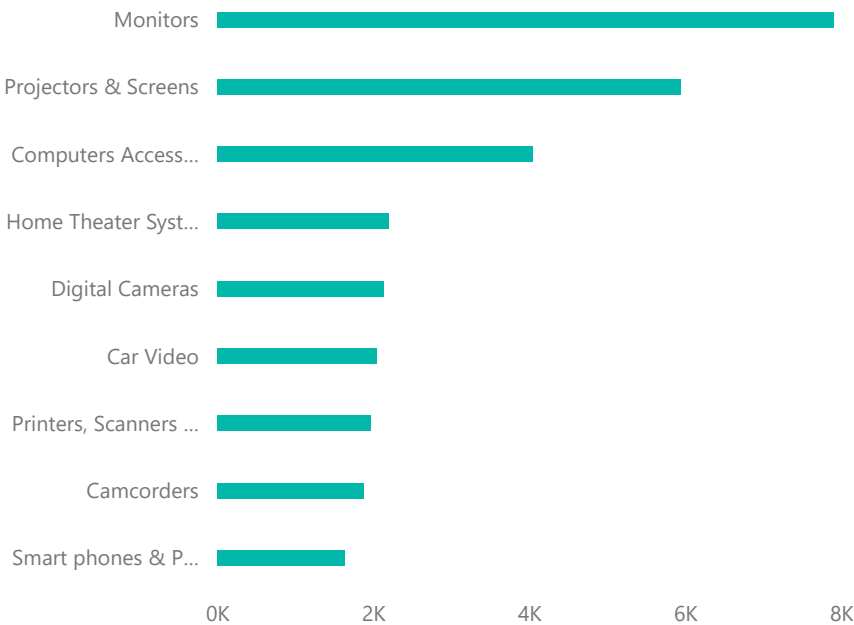
(Apr, May, Jun, Jul & Oct)

● Increase ● Decrease



Quantity Change by Top 9 Subcategories

(within previous 4 Categories)

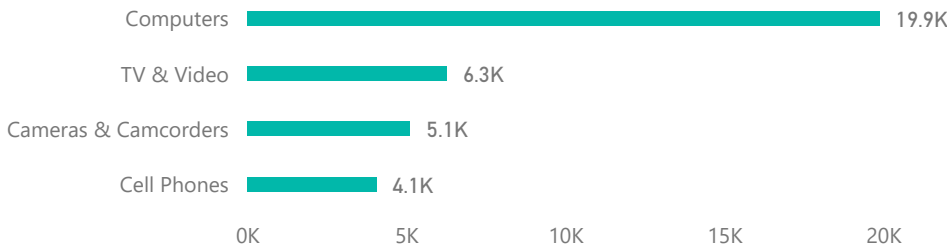


Quantity Sold %YoY -43.4% ▼

|                    |          |
|--------------------|----------|
| Quantity Sold 2011 | 431,750  |
| Quantity Sold 2012 | 244,320  |
| Quantity Sold      | -187,430 |

Quantity Sold Recovery by Category

(within Economy)

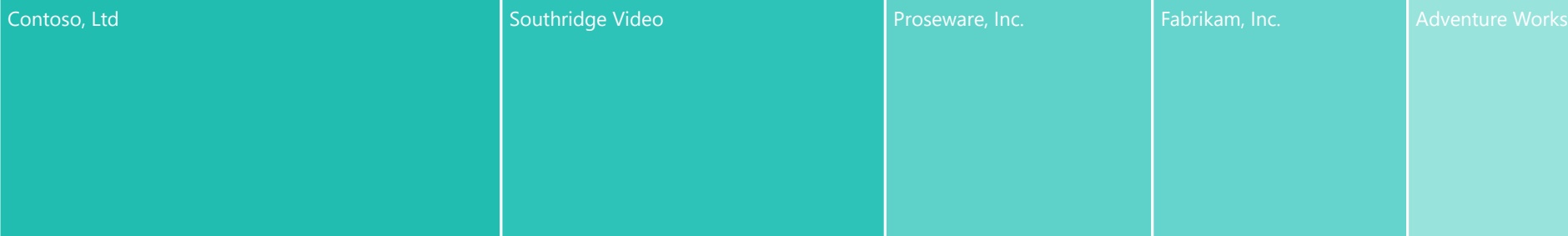


Avg. Selling Price %YoY -4.4% ▼

|            |          |
|------------|----------|
| ASP 2011   | \$334.6  |
| ASP 2012   | \$319.9  |
| ASP Change | (\$14.6) |

Quantity Change by Top 5 Manufacturers

(within previous 9 Subcategories)



# Recommendations

IntroductionExecutive OverviewRoot-cause AnalysisFinancial ImpactRecovery AnalysisRecommendations

CriticalHighLow

**i** Limited available data. Some recommendations require further investigation & validation before action

Revenue %YoY-43.1% ▼

Revenue 2011\$152.2M  
Revenue 2012\$86.6M  
Revenue Change-\$65.6M

Gross Profit %YoY-43.7% ▼

Profit 2011\$89.3M  
Profit 2012\$50.3M  
Profit Change-\$39.0M

Quantity Sold %YoY-41.9% ▼

Quantity 2011442,660  
Quantity 2012257,330  
Quantity Change-185,330

## ROOT-CAUSE ANALYSIS

| Key Drivers                                                                                            | Period                                                                                             | Product                                                                                                       | Manufacturer                                                                        | Geography                                                                       |
|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 41.9% decline in Quantity Sold contributed to the 43.1% revenue decline.                               | 185K unit decline concentrated in Apr, May, Jun, Jul & Oct.                                        | 103% of the quantity decline came from Regular class. 98.8% from 5 categories and 86.7% from 9 subcategories. | 79% of the decline is concentrated among 4 manufacturers.                           | 90% decline came from North America. Within it, 94.1% decline came from the US. |
| Prioritize customer behavior, competition, macroeconomic factors, inventory, marketing, & product mix. | Investigate what changed during these months and identify the factors contributing to the decline. | Prioritize affected products for targeted recovery actions.                                                   | Review supplier performance across pricing, quality, availability, and reliability. | Prioritize North America, particularly the US for recovery                      |

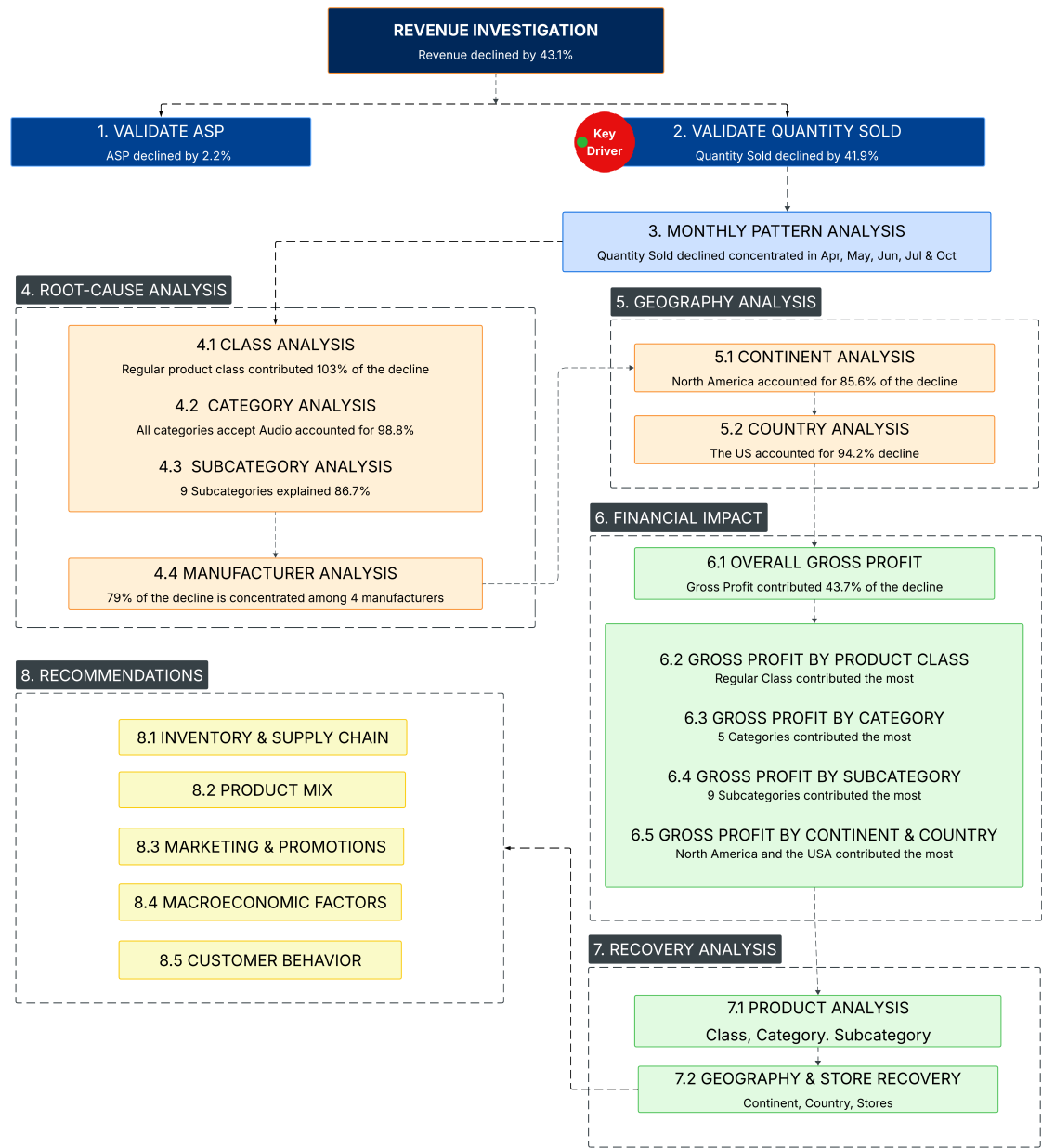
## FINANCIAL IMPACT

| Gross Profit                                                                                                 | Revenue Recovery                                                                              |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 43.7% gross profit decline occurred alongside the 43.1% revenue decline.                                     | Prioritize revenue recovery as the primary path to restoring gross profit.                    |
| Cost & Margin                                                                                                | Cost & Margin                                                                                 |
| 42.4% cost decline did not offset the revenue loss, while profit margin declined 42.6%.                      | Maintain cost controls and monitor margin recovery alongside revenue recovery                 |
| All Dimensions                                                                                               | All Dimensions                                                                                |
| All product categories, classes, manufacturers, continents, and countries contributed to the profit decline. | Focus recovery efforts on identified revenue drivers rather than broad cost-cutting measures. |

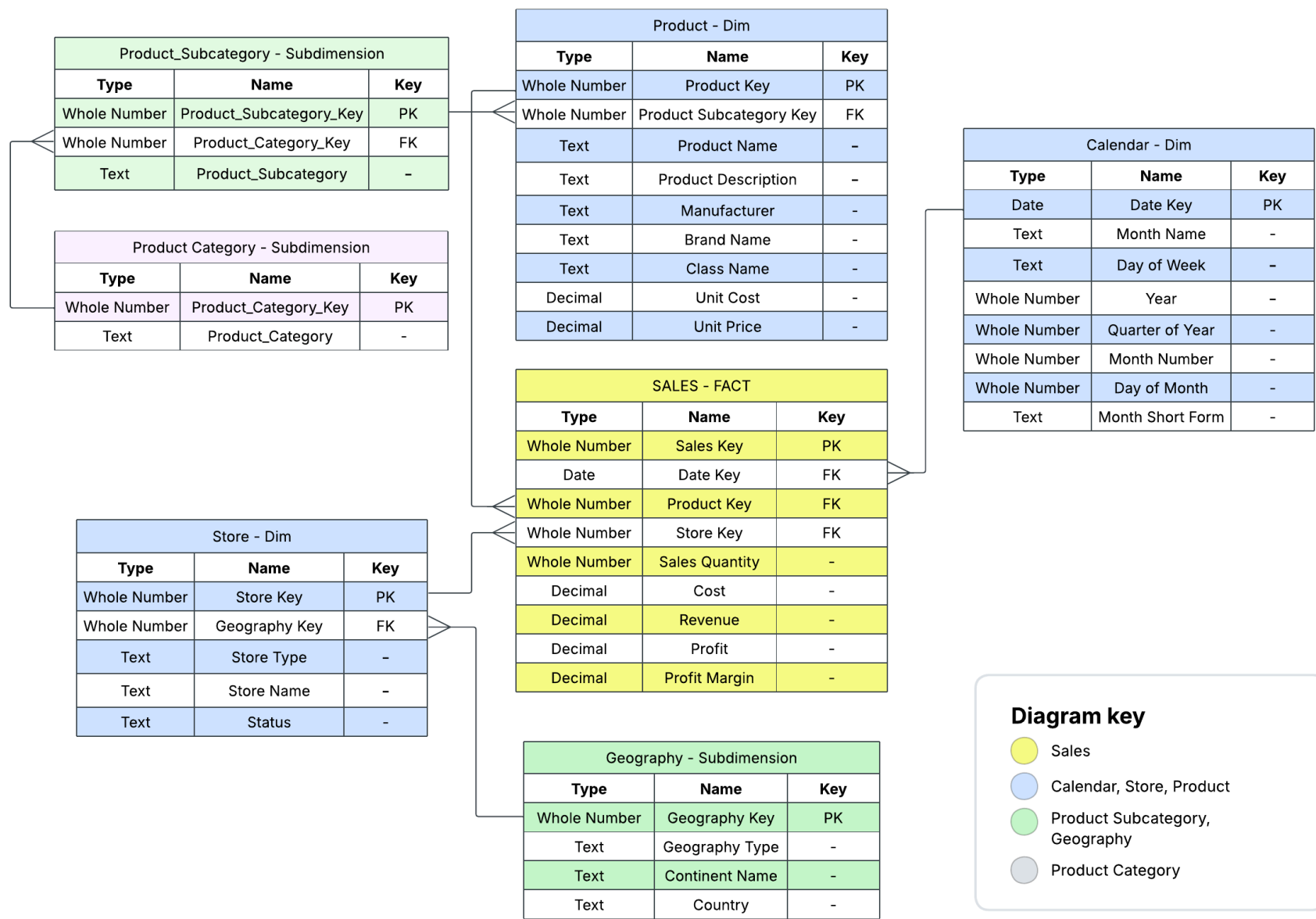
## RECOVERY ANALYSIS

| Recovery Performance                                                                                                | Recovery Action                                                                           |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 79.9% revenue decline, 43.4% quantity decline, and 4.4% ASP decline during the recovery period.                     | Apply root-cause actions for quantity decline & review pricing opportunities              |
| Product & Supplier                                                                                                  | Product & Supplier                                                                        |
| 35.4K units recovered across 4 categories & related subcategories. Top 5 manufacturers contributed to the recovery. | Identify recovery drivers and strengthen partnerships with high-performing manufacturers. |
| Geography                                                                                                           | Geography                                                                                 |
| Asia was the top recovering continent, led by Pakistan, Armenia, China, and Japan.                                  | Analyze recovery drivers in these markets and replicate successful approaches.            |

# ANALYTICAL WORKFLOW

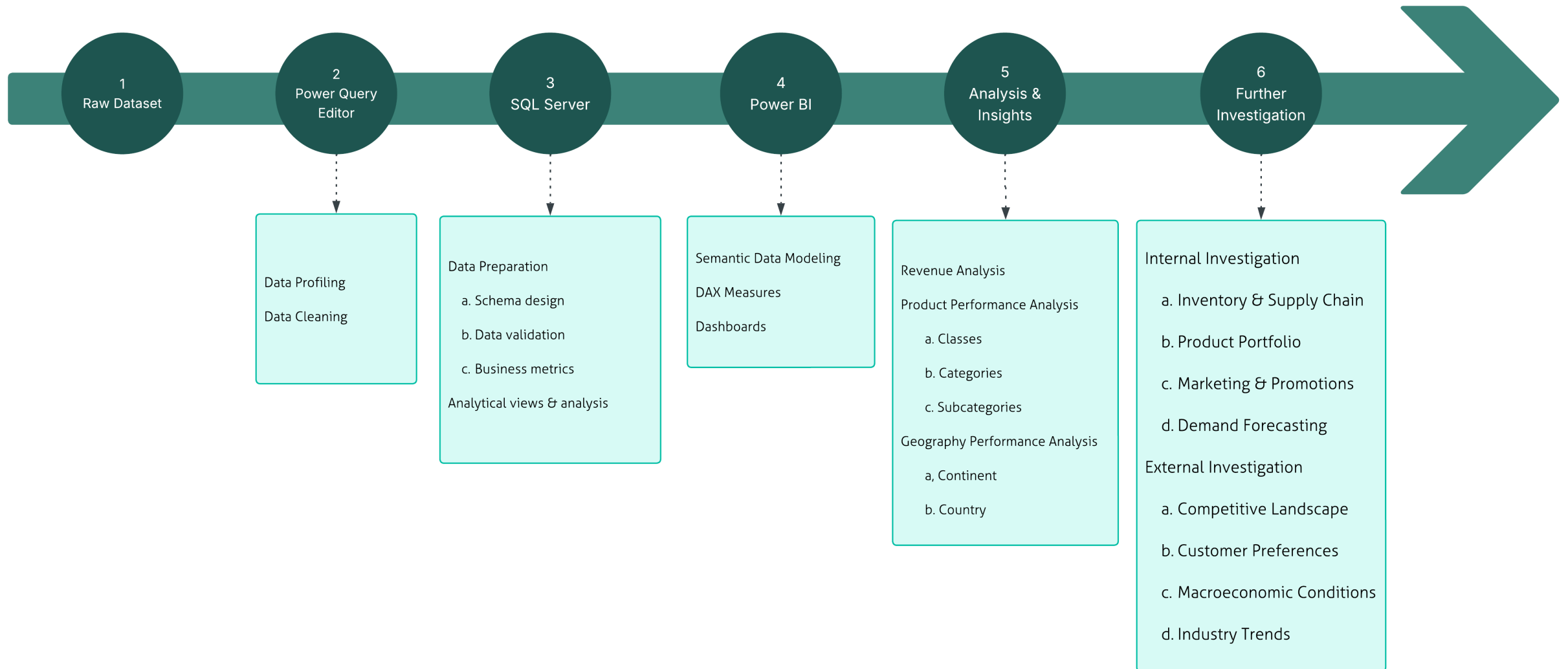


# ENTITY RELATIONSHIP DIAGRAM





# PROJECT WORKFLOW



# DATA DICTIONARY

| Table Name          | Column Name             | Description                                                              | Data Type          | Key |
|---------------------|-------------------------|--------------------------------------------------------------------------|--------------------|-----|
| Sales               | Sales_Key               | Unique identifier for each sales transaction                             | Whole Number       | PK  |
| Sales               | Date_Key                | Links each sales transaction to the Calendar table                       | Date               | FK  |
| Sales               | Store_Key               | Links each sales transaction to the Store table                          | Whole Number       | FK  |
| Sales               | Product_Key             | Links each sales transaction to the Product table                        | Whole Number       | FK  |
| Sales               | Sales_Quantity          | Number of units sold in each sales transaction                           | Whole Number       | -   |
| Sales               | Revenue                 | Total Revenue generated from the sales transaction                       | Decimal Number     | -   |
| Sales               | Profit                  | Total Profit earned from the sales transaction                           | Decimal Number     | -   |
| Sales               | Profit_Margin           | Profit expressed as a percentage of revenue for a sales transaction      | Decimal Number (%) | -   |
| Product             | Product_Key             | Unique identified for each product                                       | Whole Number       | PK  |
| Product             | Product_Subcategory_Key | Links each product to its corresponding product subcategory              | Whole Number       | FK  |
| Product             | Product_Name            | Name of the Product                                                      | Text               | -   |
| Product             | Product_Description     | Detailed description of the product                                      | Text               | -   |
| Product             | Manufacturer            | Name of the product manufacturer                                         | Text               | -   |
| Product             | Brand_Name              | Brand under which the product is sold                                    | Text               | -   |
| Product             | Class_Name              | Product classification                                                   | Text               | -   |
| Product             | Unit_Cost               | Cost incurred per unit of the product                                    | Decimal Number     | -   |
| Product             | Unit_Price              | Selling Price per unit of the product                                    | Decimal Number     | -   |
| Product_Category    | Product_Category_Key    | Unique identifier for each product category.                             | Whole Number       | PK  |
| Product_Category    | Product_Category        | Name of the product category                                             | Text               | -   |
| Product_Subcategory | Product_Subcategory_Key | Unique identifier for each product subcategory.                          | Whole Number       | PK  |
| Product_Subcategory | Product_Category_Key    | Name of the product subcategory                                          | Whole Number       | FK  |
| Product_Subcategory | Product_Subcategory     | Name of the product subcategory                                          | Text               | -   |
| Stores              | Store_Key               | Unique identifier for each store.                                        | Whole Number       | PK  |
| Stores              | Geography_Key           | Links each store to its corresponding geographic location.               | Whole Number       | FK  |
| Stores              | Store_Type              | Type or format of the store                                              | Text               | -   |
| Stores              | Store_Name              | Name of the store.                                                       | Text               | -   |
| Stores              | Status                  | Operational status of the store (e.g., On or Off).                       | Text               | -   |
| Geography           | Geography_Key           | Unique identifier for each geographic location.                          | Whole Number       | PK  |
| Geography           | Geography_Type          | Classification of the geographic level (e.g., City, Country, Continent). | Text               | -   |
| Geography           | Continent_Name          | Name of the continent where the location is situated.                    | Text               | -   |
| Geography           | Country                 | Name of the country associated with the geographic location.             | Text               | -   |
| Calendar            | Date_Key                | Unique date identifier used to link transactions to the calendar.        | Date               |     |
| Calendar            | Month_Name              | Full name of the month (e.g., April, May).                               | Text               |     |
| Calendar            | Day_Of_Week_Name        | Name of the day of the week (e.g., Monday, Tuesday).                     | Text               |     |
| Calendar            | Year                    | Calendar year of the date.                                               | Whole Number       |     |
| Calendar            | Quarter_Of_Year         | Quarter of the year (Q1-Q4).                                             | Whole Number       |     |
| Calendar            | Month_Of_Year           | Numeric representation of the month (1-12).                              | Whole Number       |     |
| Calendar            | Day_Of_Month            | Day of the month (1-31).                                                 | Whole Number       |     |